Quantitative Report & Regression Results

This report presents a comprehensive summary of the regression analyses performed on the Banks' financial and ESG data. All models are multiple linear regressions in which key financial performance metrics serve as dependent variables, while the 10 ESG sub-scores (Emission, Resource Use, Innovation, Human Rights, Product Responsibility, Workforce, Community, Management, Shareholders, CSR Strategy) act as independent variables.

The analysis covers the full sample as well as group-specific regressions based on banks` assets and ESG-driven clustering. Model fit is evaluated through Multiple R, R Square, Adjusted R Square, Standard Error, F-statistic and Significance F, while coefficient tables provide detailed parameter estimates including betas, standard errors, t-statistics, p-values and 95% confidence intervals.

The report is structured to describe the model setup, present the main regression results, summarize subgroup outcomes, outline the clustering logic, review key variable statistics, and assess multicollinearity through Variance Inflation Factors (VIF).

Model Structure Explanation

All regression models implemented in this analysis are Ordinary Least Squares (OLS) multiple linear regressions. The dependent variables consist of standard banking performance metrics, including Return on Assets (ROA), Return on Equity (ROE), Net Profit Margin, OPEX, Interest Cost Income ratio, Cost Income Ratio, Interest Coverage Ratio, Asset Turnover, Liquidity Coverage Ratio, Leverage Ratio, Debt to Equity, and Risk-Weighted Assets (RWA).

Each model uses the 10 ESG sub-scores—Emission, Resource Use, Innovation, Human Rights, Product Responsibility, Workforce, Community, Management, Shareholders, and CSR Strategy—as independent variables. Total ESG scores are intentionally excluded because they are composite averages derived directly from these sub-scores. An intercept term is included in every model.

Model performance is evaluated using Multiple R, R Square, Adjusted R Square, Standard Error, the number of observations, the overall F-statistic, and its associated significance (Significance F). Coefficient tables report the estimated beta coefficients alongside their standard errors, t-statistics, p-values, and 95% confidence intervals, providing a complete statistical summary for each regression specification.

A standard multiple linear regression model was applied:
KPI = β_0 + β_1(Emission) + … + β_{10}(CSR Strategy) + ε
Each model:

- Uses all 10 ESG predictors
- Has N = 57 banks (unless group regressions)
- Includes intercept
- Includes full ANOVA output (F-test, significance F)
- Includes coefficient table (β, Std. Error, t-stat, p-value, CI)

Main Regression Results (ESG Scores Models)

The following pages present the detailed regression results for the models predicting the main performance metrics from the sub-ESG scores. There are 12 separate models in this analysis.

Model Fit (R^2 Values)
Across all 12 models:

- **Highest R^2:** Leverage Ratio (≈ 0.31)
- **Moderate R^2:** Interest Cost Income (≈ 0.29), Liquidity (≈ 0.26)
- **Weak R^2:** Asset Turnover (≈ 0.10), OPEX (≈ 0.14)

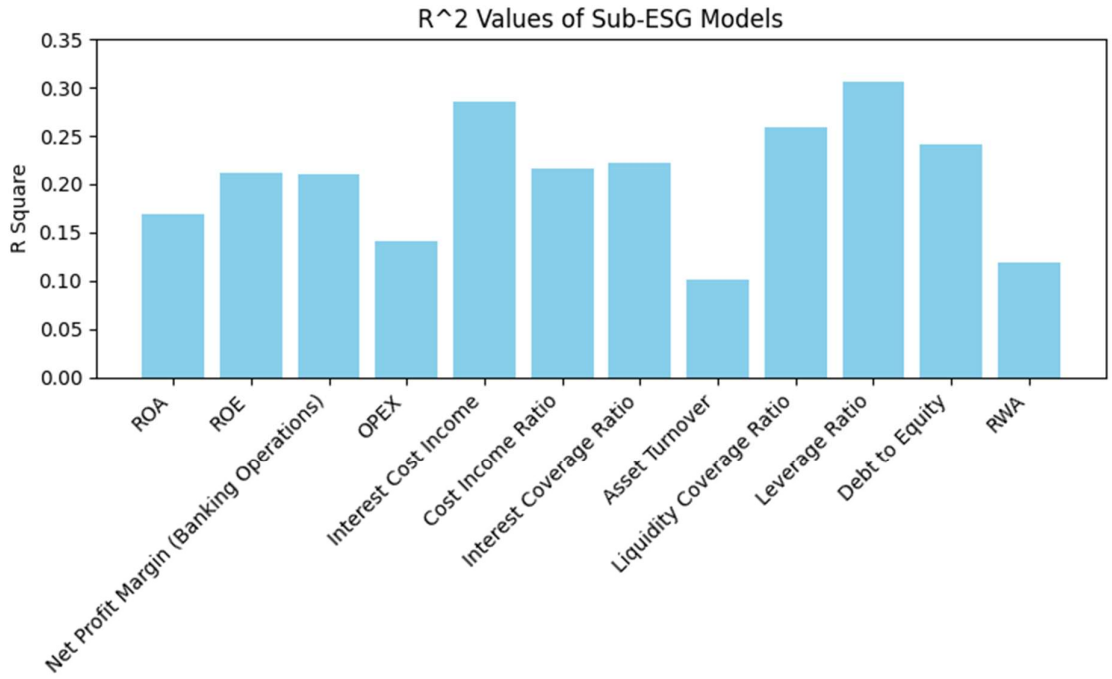
Conclusion: ESG sub-scores explain only 10–30% of KPI variance.

F-tests
Most models fail the F-test (Significance F > 0.05), meaning:
The ESG block does not jointly predict financial KPIs in a statistically significant way.

Significant ESG Predictors

Only 4 ESG variables ever reach significance ($p < 0.05$):

- Innovation → ROA
- Resource Use → Interest Cost Income
- Human Rights → Interest Cost Income
- CSR Strategy → Liquidity Coverage Ratio



Simplified Regression Results

ROA
R^2: 0.170, Adjusted R^2: -0.011, Observations: 57
F-statistic: 0.940, Significance: 0.507
Significant Variables:
Intercept: β=0.00945, p=0.025
Innovation: β=-0.00006, p=0.049

ROE
R^2: 0.212, Adjusted R^2: 0.041, Observations: 57
F-statistic: 1.240, Significance: 0.292
Net Profit Margin (Banking Operations)
R^2: 0.211, Adjusted R^2: 0.039, Observations: 57
F-statistic: 1.227, Significance: 0.299
Significant Variables:
Intercept: β=0.25484, p=0.024

OPEX
R^2: 0.141, Adjusted R^2: -0.046, Observations: 57
F-statistic: 0.755, Significance: 0.670
Significant Variables:
Intercept: β=0.99015, p=0.000

Interest Cost Income
R^2: 0.285, Adjusted R^2: 0.130, Observations: 57
F-statistic: 1.837, Significance: 0.080
Significant Variables:
Intercept: β=0.45730, p=0.005
Resource Use: β=0.00419, p=0.010
Human rights: β=-0.00257, p=0.010

Cost Income Ratio
R^2: 0.216, Adjusted R^2: 0.046, Observations: 57
F-statistic: 1.271, Significance: 0.275
Significant Variables:
Intercept: β=0.47274, p=0.003

Interest Coverage Ratio
R^2: 0.222, Adjusted R^2: 0.053, Observations: 57
F-statistic: 1.311, Significance: 0.253
Asset Turnover
R^2: 0.101, Adjusted R^2: -0.094, Observations: 57
F-statistic: 0.519, Significance: 0.868
Significant Variables:
Intercept: β=0.02675, p=0.041

Liquidity Coverage Ratio
R^2: 0.258, Adjusted R^2: 0.097, Observations: 57
F-statistic: 1.603, Significance: 0.136
Significant Variables:
CSR Strategy: β=-0.01508, p=0.015

Leverage Ratio
R^2: 0.306, Adjusted R^2: 0.156, Observations: 57
F-statistic: 2.031, Significance: 0.051
Significant Variables:
Intercept: β=0.10395, p=0.000

Debt to Equity
R^2: 0.241, Adjusted R^2: 0.076, Observations: 57
F-statistic: 1.462, Significance: 0.185
Significant Variables:
Intercept: β=96.52089, p=0.000

RWA
R^2: 0.119, Adjusted R^2: -0.073, Observations: 57
F-statistic: 0.618, Significance: 0.790
Significant Variables:
Intercept: β=0.48426, p=0.002

Group Analysis

Group 1 (High Asset Banks) – Regression Analysis Summary

Group 1 consists of banks with higher total assets. An examination of all 12 regression models yields the following insights:
• The R^2 values range between approximately 0.18 and 0.60, indicating **low to moderate explanatory power**.
• **None of the models pass the F-test** (all Significance F values are greater than 0.05), meaning the ESG score block does not jointly explain variations in the financial KPIs.
• Only a few ESG variables appear as individually significant in isolated cases:
– **Workforce** (significant in two models)
– **Community** (significant in one model)
These occurrences are sporadic and do not form any consistent or economically interpretable pattern.
Conclusion:
Among banks with large asset bases, ESG scores show **no statistically meaningful ability** to explain profitability, efficiency, liquidity, or structural KPIs. The models remain weak, and ESG factors do not emerge as reliable predictors within this subgroup.

Group 2 (Low Asset Banks) – Regression Analysis Summary

Group 2 is composed of banks with lower total asset levels. This subgroup includes nearly the same number of regression models (one appears to be missing in the dataset), and the findings closely mirror those of Group 1:

• The R^2 values fall between roughly 0.15 and 0.57, again indicating **low to moderate explanatory strength**.

• **No model achieves joint significance** (all Significance F values remain above 0.14).

• A small number of ESG variables reach individual significance:

– **Emission** (significant in two models)

– **Resource Use** (significant in one model)

– **Workforce** (significant in one model)

Similar to Group 1, these significant results are isolated and inconsistent, showing no stable relationship between ESG sub-scores and financial KPIs.

Conclusion:

For banks with smaller asset bases as well, ESG indicators **do not meaningfully explain** financial performance metrics. The patterns are weak, inconsistent, and statistically unsupported.

Cluster Analysis

The dataset has been partitioned into two groups through cluster analysis based on the ESG sub-score components. The *Cluster Assignments* sheet provides the cluster label and descriptive classification for each bank, identifying a **High ESG** cluster with 48 banks and a **Low ESG** cluster with 9 banks.

Using ESG sub-dimensions → k-means → 2 clusters:

Cluster Sizes

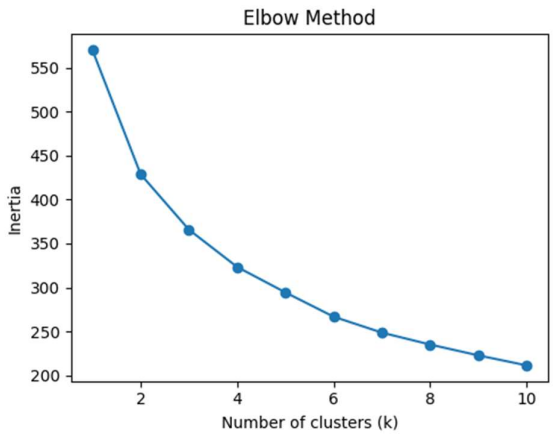
- High ESG: 48 banks
- Low ESG: 9 banks

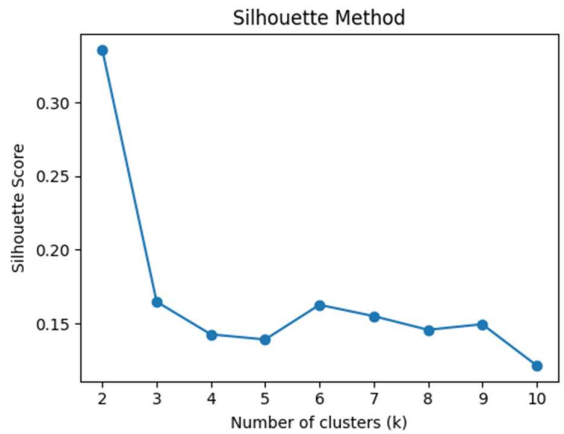
Cluster Differences (using KPI Comparison tab)

Only 3 KPIs significantly differ ($p < 0.05$):

1. Assets — High ESG banks are ~10x larger
2. 1–Debt-to-Equity — High ESG banks have slightly lower leverage
3. Leverage Ratio — High ESG banks use less leverage

All other KPIs (ROA, ROE, NPM, Liquidity, Asset Turnover, etc.) show no statistical difference.





Variable	High ESG Mean	High ESG Std	Low ESG Mean	Low ESG Std	Difference	PValue
Assets in Euro	563601.94	685929.25	56812.06	49945.68	506789.88	0.000006
1-Debt to Equity	84.254308	3.899793	87.006326	3.298443	-2.752018	0.044782
Interest Coverage	2.902471	5.117148	2.778483	3.290526	0.123988	0.926438
1-Interest Cost Income	0.666943	0.153249	0.708455	0.211989	-0.041511	0.587859
1-Cost Income Ratio	0.422814	0.132841	0.388700	0.250338	0.034114	0.699740
Liquidity Coverage	1.840229	0.962785	1.807778	0.232475	0.032451	0.839193
1-RWA	0.666448	0.122058	0.639721	0.218756	0.026726	0.729888
ROE	0.100603	0.052887	0.075568	0.053744	0.025035	0.224729
Levarage Ratio	0.056090	0.012185	0.068889	0.016011	-0.012799	0.046442
Asset Turnover	0.025055	0.012511	0.021437	0.008886	0.003618	0.313837
Net Profit Margin	0.207847	0.104953	0.206811	0.141450	0.001036	0.983735
1-OPEX	0.986011	0.006504	0.986975	0.007411	-0.000964	0.722560
ROA	0.006567	0.003986	0.005911	0.004384	0.000657	0.684183

Variable	Cluster 0 (48 Banks)	Cluster 1 (9 Banks)
Emission	83.21	57.33
Resource Use	84.71	56.11
Innovation	75.58	43.11
Human Rights	78.81	35.22
Product Responsibility	63.42	45.78
Workforce	85.25	67
Community	69.63	34.33
Management	73.83	50.56
Shareholders	60.83	48.56
CSR Strategy	77.58	31.78

Descriptive Statistics

ESG Variables Summary

- Average ESG scores mostly between 60–80
- Innovation & Human Rights show the highest variability
- Workforce typically scores higher across banks
- CSR Strategy varies meaningfully across banks (indicative of policy maturity)

Score	Average	Highest	Lowest	ST
Total	70.78947368	92	27	13.67
Environmental	73.52631579	98	19	19.69
Emission	79.12280702	100	39	14.79
Innovation	70.45614035	98	5	25.82
Resource Use	80.19298246	100	6	17.52
Social	71.9122807	95	16	14.94
Human Rights	71.92982456	96	0	26.93
Product Responsibility	60.63157895	100	0	27.58
Workforce	82.36842105	100	27	13.4
Community	64.05263158	99	3	24.57
Governance	68.14035088	94	25	18.86
Management	70.15789474	100	3	23.49
Shareholders	58.89473684	98	3	26.06
CSR Strategy	70.35087719	100	3	27.07

Financial Variables Summary

- ROA and ROE show typical European bank levels
- Leverage ratios display meaningful spread
- Liquidity and RWA vary significantly between business models

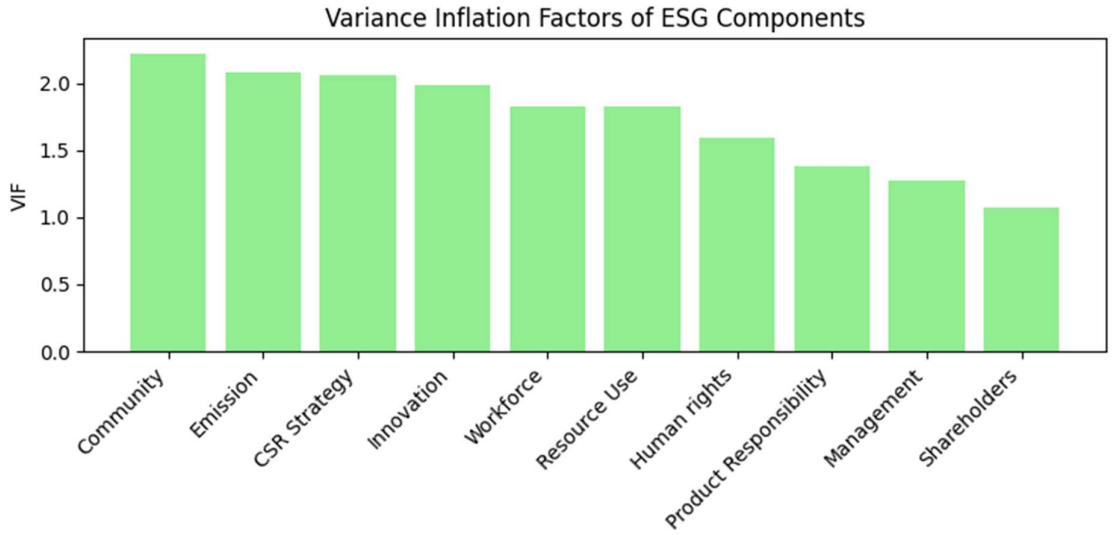
Financial Ratio	Average	Highest	Lowest	ST
ROA	0.0063	0.019983821	-0.00158115	0.004017265
ROE	0.0949	0.235423542	-0.030573863	0.053338556
RWA	0.6807	0.980357354	0.261828956	0.152988347
OPEX	0.9862	0.995850968	0.962208791	0.006593564
Net Profit Margin	0.2077	0.481052632	-0.054647736	0.110015351
Liquidity Coverage Ratio	1.84	7.86	1.29	0.886477533
Leverage Ratio	0.059	0.109	0.0403	0.008600664
Interest Coverage Ratio	2.883	32.10526316	-1.087947883	4.850334358
Interest cost income	0.6735	0.946778711	0.220677672	0.162370228
Debt to Equity	0.85	0.931631464	0.754183755	4.12976556
Cost Income	0.4174	0.646871686	-0.259115593	0.154664147
Asset Turnover	0.0245	0.09050442	0.0102	0.012017782

Multicollinearity Diagnostics: VIF Results

The Variance Inflation Factor (VIF) analysis returns the following values for the set of ESG sub-dimensions included in the regression models. All VIF values are below the threshold of 5 indicating that multicollinearity is acceptably low across the predictors.

VIF Results (All < 2.3)
The VIF table shows:

- Minimum: Shareholders (≈ 1.07)
- Maximum: CSR Strategy (≈ 2.22)
- All < 5 → no multicollinearity concern

This validates the stability of the regression coefficients.

Variable	VIF
Community	2.221297232
Emission	2.080320465
CSR Strategy	2.055702422
Innovation	1.984137852
Workforce	1.828491616
Resource Use	1.824478736
Human rights	1.59285859
Product Responsibility	1.381915203
Management	1.272539535
Shareholders	1.07483701